

Carwash: Smart Car Wash Management System

Software Requirement Specification

Project Code: Carwash

Document Code: SRS-CW-V1.0

FPT University, July 2026

Record of change

*A - Added M - Modified D - Deleted

Effective Date	Changed Items	A/M/D	Change Description	New Version
21/07/2026	Full document	A	Create Carwash SRS based on provided template.	V1.0

SIGNATURE PAGE

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1. Introduction

This Software Requirements Specification (SRS) describes the Carwash system, a web-based platform for booking car wash services, managing vehicles, assigning staff, tracking wash sessions, processing payments, applying promotions, and monitoring business operations.

1.1 Purpose

The purpose of this document is to define the main functional and non-functional requirements for Carwash. It is used as a shared reference for the project team, testers, reviewers, and stakeholders.

1.2 Scope

Carwash supports customers, staff, managers, and administrators. Customers can register, manage vehicles, book services, pay online, receive notifications, and review service history. Staff can check in vehicles and update wash progress. Managers can monitor the operations queue, assign staff, view reports, and configure operational settings. Administrators can manage master data, users, promotions, and system reports.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
SRS	Software Requirement Specification
Customer	End user who books and pays for car wash services
Staff	Employee who handles vehicle check-in and wash tasks
Manager	Branch/operator who supervises daily operations
Admin	System-level role that manages configuration and data
VNPay	Payment gateway used for online payment

1.4 References

- IEEE 830-1998 Recommended Practice for Software Requirements Specifications.
- Carwash source code and project requirements.
- Team UI/API implementation notes.

1.5 Overview

The remaining sections describe the product context, major user roles, functional use cases, non-functional requirements, and supporting assumptions.

2. Overall Description

2.1 Product Perspective

Carwash is a full-stack web application consisting of a responsive frontend and a backend API. It centralizes the customer booking journey and the internal operation flow so that every booking can be tracked from reservation to completion.

2.2 Product Functions

- Provide customer account, vehicle, booking, payment, loyalty, and notification features.
- Provide staff screens for check-in, task progress, and session handling.
- Provide manager screens for operations queue, staff workload, reports, settings, and notifications.
- Provide admin screens for accounts, services, promotions, discounts, packages, reviews, bookings, and reporting.

2.3 User Classes and Characteristics

User Class	Main Responsibility
Customer	Books services and follows wash progress.
Staff	Executes assigned wash sessions and updates status.
Manager	Supervises branch operations and handles exceptions.
Admin	Controls system data, users, and business configuration.

2.4 Operating Environment

The system runs in a modern web browser. The frontend communicates with the backend through REST APIs. The backend stores operational data in a database and integrates with payment and notification services.

2.5 Assumptions and Dependencies

- Internet connection is available for users and staff devices.
- Backend APIs are reachable and protected by authentication tokens.
- Payment processing depends on the configured VNPay sandbox or production environment.

3. Functional Requirements

3.1 Use Case Summary

ID	Use Case	Primary Actor	Priority
UC001	Register and Login	Customer / Staff / Manager / Admin	High
UC002	Manage Vehicles	Customer	High
UC003	Book Car Wash Service	Customer	High
UC004	Track Booking and History	Customer	Normal
UC005	Process Payment	Customer	High
UC006	Manage Operations Queue	Manager	High
UC007	Check-in Vehicle	Staff	High
UC008	Update Wash Session	Staff	High
UC009	Manage Staff and Workload	Manager	High
UC010	Manage Services and Promotions	Admin / Manager	Normal
UC011	View Reports	Manager / Admin	Normal
UC012	Manage Notifications	Customer / Manager	Normal

UC001 - Register and Login

UC001 SPECIFICATION			
Use-case Name	Register and Login	Priority	High
Actors	All users	Version	1.0
Brief Description	User creates an account or signs in to access the correct workspace.	Pre-condition	User has valid credentials or registration information.
Main Flow	Enter email/phone and password, verify OTP when required, then redirect by role.	Post-condition	Authenticated session is created.

Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline
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UC002 - Manage Vehicles

UC002 SPECIFICATION			
Use-case Name	Manage Vehicles	Priority	High
Actors	Customer	Version	1.0
Brief Description	Customer adds, edits, or removes vehicle information for future bookings.	Pre-condition	Customer is logged in.
Main Flow	Open vehicle page, enter vehicle plate and details, save changes.	Post-condition	Vehicle information is stored and available for booking.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

UC003 - Book Car Wash Service

UC003 SPECIFICATION			
Use-case Name	Book Car Wash Service	Priority	High
Actors	Customer	Version	1.0
Brief Description	Customer selects service, date/time, vehicle, branch/session information, and confirms a booking.	Pre-condition	Customer is logged in and has at least one vehicle.
Main Flow	Choose service, choose schedule, review price, apply discount if available, confirm booking.	Post-condition	Booking is created with initial status.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

UC004 - Track Booking and History

UC004 SPECIFICATION			
Use-case Name	Track Booking and History	Priority	Normal
Actors	Customer	Version	1.0
Brief Description	Customer views upcoming, active, and completed bookings.	Pre-condition	Customer has booking records.
Main Flow	Open booking/history page,	Post-condition	Booking detail is

	filter by status, view detail or wash timeline.		displayed.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

UC005 - Process Payment

UC005 SPECIFICATION			
Use-case Name	Process Payment	Priority	High
Actors	Customer	Version	1.0
Brief Description	Customer pays for eligible booking using supported payment method.	Pre-condition	Booking requires payment and payment gateway is available.
Main Flow	Choose payment method, submit payment, return to payment result page.	Post-condition	Payment status is updated.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

UC006 - Manage Operations Queue

UC006 SPECIFICATION			
Use-case Name	Manage Operations Queue	Priority	High
Actors	Manager	Version	1.0
Brief Description	Manager monitors check-in, wash progress, delays, and operational exceptions.	Pre-condition	Manager is logged in.
Main Flow	Open operations queue, filter sessions, review alerts, assign or prioritize tasks.	Post-condition	Queue state is updated for staff execution.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

UC007 - Check-in Vehicle

UC007 SPECIFICATION			
Use-case Name	Check-in Vehicle	Priority	High
Actors	Staff	Version	1.0
Brief Description	Staff confirms customer arrival and starts the operational flow.	Pre-condition	Booking exists and is not completed or cancelled.
Main Flow	Search booking, verify	Post-condition	Booking moves to

	vehicle/customer, click Check-in.		checked-in stage.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

UC008 - Update Wash Session

UC008 SPECIFICATION			
Use-case Name	Update Wash Session	Priority	High
Actors	Staff	Version	1.0
Brief Description	Staff updates session from waiting to washing, inspection, and completion.	Pre-condition	Session is assigned or available to staff.
Main Flow	Open assigned task, update progress status, add notes when needed.	Post-condition	Session timeline reflects the latest status.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

UC009 - Manage Staff and Workload

UC009 SPECIFICATION			
Use-case Name	Manage Staff and Workload	Priority	High
Actors	Manager	Version	1.0
Brief Description	Manager views staff workload, KPI, availability, and assigns work.	Pre-condition	Manager has permission to manage staff.
Main Flow	Open staff page, filter staff, view performance, assign suitable staff.	Post-condition	Assignment or staff status is updated.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

UC010 - Manage Services and Promotions

UC010 SPECIFICATION			
Use-case Name	Manage Services and Promotions	Priority	Normal
Actors	Admin / Manager	Version	1.0
Brief Description	Authorized user configures service catalog, discounts, vouchers, or promotions.	Pre-condition	User has management permission.

Main Flow	Create or edit service/promotion information and save.	Post-condition	Catalog or promotion data becomes available to users.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

UC011 - View Reports

UC011 SPECIFICATION			
Use-case Name	View Reports	Priority	Normal
Actors	Manager / Admin	Version	1.0
Brief Description	Manager/Admin views revenue, bookings, staff KPI, and service quality reports.	Pre-condition	Report data is available from backend.
Main Flow	Choose time range and filters, view cards/charts/tables, export when needed.	Post-condition	Report is displayed or exported.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

UC012 - Manage Notifications

UC012 SPECIFICATION			
Use-case Name	Manage Notifications	Priority	Normal
Actors	Customer / Manager	Version	1.0
Brief Description	User receives, reads, and manages notifications related to bookings or operations.	Pre-condition	User is logged in.
Main Flow	Open notification page, filter/read notification, mark as read or clear items.	Post-condition	Notification state is updated.
Exception	Invalid input, unavailable slot, failed payment, or unauthorized action is shown with a clear message.	Status	Baseline

4. Non-functional Requirements

4.1 Usability

- The interface should be clear, responsive, and consistent across customer, staff, manager, and admin pages.
- Common workflows should require minimal steps and show meaningful loading, empty, and error states.

4.2 Reliability

- The system should preserve booking/payment state and avoid duplicate processing where possible.
- Completed bookings must not be reassigned to other active tasks.

4.3 Performance

- Main pages should load quickly under normal demo data volume.
- Filters, charts, and queue updates should respond without blocking the interface.

4.4 Security

- Role-based access control must protect customer, staff, manager, and admin workspaces.
- Sensitive payment configuration and tokens must be stored outside frontend source code.

4.5 Maintainability

- Frontend components should reuse shared API services, hooks, layout, and UI patterns.
- Backend APIs should use consistent response shapes and validation rules.

4.6 Compatibility

- The system should support current versions of Chrome, Edge, and common desktop/mobile browsers.

5. Supporting Information

5.1 Data Entities

Entity	Short Description
User	Account profile and role information
Vehicle	Customer vehicle information
Booking	Reservation, service, schedule, and payment state
Wash Session	Operational tracking of a booking after check-in
Service	Available wash/detailing package
Promotion / Discount	Campaign or voucher applied to eligible bookings
Notification	Message shown to users or managers
Review	Customer feedback and rating

5.2 Constraints

- This SRS intentionally focuses on the main demo-ready requirements and avoids deep technical implementation details.
- API names, database fields, and UI wording may be refined during implementation as long as the core behavior remains unchanged.

5.3 Approval

This document is approved as the baseline SRS for the Carwash project version 1.0.